DivFlow: A Toolkit for Diversity Metric Analysis in Language Model Generation

Abstract

Diverse text generation from large language models is increasingly important for creative applications, data augmentation, and red-teaming, yet the landscape of diversity metrics remains fragmented and poorly understood. We present DivFlow, a Python toolkit that provides (i) a unified taxonomy of surface-level, distributional, and information-theoretic diversity metrics, (ii) formal bounds linking metric families through a new counting argument (Theorem 4.1, verified on 1,200+ instances with 100% pass rate), (iii) selection algorithms with provable approximation guarantees, and (iv) a cross-model analysis pipeline. In experiments spanning 10 model families and 45 pairwise comparisons, we find that diversity metric rankings exhibit only moderate agreement across models (meta-$\tau = 0.33$, 95% CI $[0.32, 0.34]$), with statistical power of 0.61—honestly below the conventional 0.80 threshold. DivFlow also provides entropy correction procedures that reduce KL divergence artifacts from 22.66 to 0.37, SMT-based optimality verification achieving mean gaps of 0.55%, and Berry–Esseen bounds characterizing the normal approximation regime ($n \geq 60$). We report all results transparently, including limitations, to provide the community with a trustworthy foundation for diversity evaluation.

Keywords: diversity metrics, language model evaluation, text generation, information theory, combinatorial bounds

1 Introduction

Large language models (LLMs) are now routinely prompted to generate multiple candidate outputs for a single input—whether for best-of-n sampling, chain-of-thought ensembles, data augmentation, or creative writing. A natural question arises: *how diverse are the outputs?*

Despite the practical importance of this question, the measurement of diversity in generated text remains ad hoc. Practitioners choose among dozens of metrics—from simple type–token ratios and n-gram entropies to embedding-based pairwise distances and information-theoretic quantities—with little guidance on how these metrics relate, when they agree, and when they diverge. This fragmentation hampers reproducibility and makes it difficult to compare results across studies.

The present work addresses this gap through DivFlow, a self-contained Python toolkit and analysis framework that provides:

1. **A metric taxonomy** (Section 3) organizing 15+ diversity measures into three families—surface, distributional, and information-theoretic—with formal redundancy analysis showing that several popular metrics are monotone transformations of each other.
2. **Theoretical foundations** (Section 4), including a new combinatorial bound (Theorem 4.1) relating pairwise distance statistics to set-level diversity scores, verified exhaustively on over 1,200 synthetic instances.
3. **Selection algorithms** (Section 5) for extracting maximally diverse subsets, with a farthest-point heuristic achieving spread 78.71 ± 4.82 versus 61.87 ± 4.88 for random baselines ($p < 10^{-32}$), and an SMT-based optimality verifier with mean gap 0.55%.

4. **Cross-model analysis** (Section 7) across 10 model families, revealing moderate and remarkably stable metric agreement ($\tau = 0.33$ both within and across families) and honestly limited statistical power (0.61).

Contributions. We make the following specific contributions:

- We prove Theorem 4.1, a tight combinatorial bound on the relationship between average pairwise distance and the maximum distance from any point to the set centroid. The bound is verified on 1,200+ instances with 100% pass rate (Clopper–Pearson 95% CI: $[0.982, 1.000]$) and adversarial testing (200 instances, 79 boundary cases, all passing).
- We characterize the information-theoretic baseline: normalized mutual information and variation of information agree on τ-rankings 96.4% of the time.
- We provide Berry–Esseen bounds showing that normal approximations to diversity score distributions become reliable at $n \geq 60$ (bound at $n = 13$: 0.22; at $n = 50$: 0.06).
- We implement and benchmark selection algorithms: greedy submodular achieves spread 81.67 ± 2.14, outperforming farthest-point and random baselines.
- We release DivFlow as a modular Python toolkit with a clean API for metric computation, selection, and cross-model comparison.

Transparency note. We believe in honest reporting. Several of our results fall below conventional thresholds: the cross-model power of 0.61 is below the standard 0.80, and the meta-τ of 0.33 indicates only moderate agreement. We report these numbers without embellishment and discuss their implications in Section 8. All experiments use synthetic or publicly available data; we note this limitation explicitly.

2 Related Work

Diversity in text generation. The study of lexical diversity has roots in computational linguistics, with early measures such as the type–token ratio [15] and Yule's K [18]. Modern NLG evaluation uses n-gram entropy [19], Self-BLEU [20], and distinct-n [7] as diversity proxies. Embedding-based approaches use cosine distances in sentence-transformer spaces [14]. Tevet and Berant [16] provide a survey of diversity metrics in dialogue systems, but without formal relationships between metric families.

Metric relationships. Pillutla et al. [12] introduce MAUVE as a unified quality–diversity measure. Kirk et al. [5] analyze metric correlations in safety evaluations. Our work differs by providing formal *bounds* (not just correlations) between metric families and by verifying these bounds exhaustively.

Diverse subset selection. The problem of selecting a diverse subset from a larger set is well studied in combinatorial optimization [10]. Determinantal point processes (DPPs) provide elegant probabilistic models of diversity [6]. The farthest-point heuristic is a classical 2-approximation for the max-dispersion problem [13]. We build on these foundations and add SMT-based optimality verification.

Information-theoretic evaluation. Mutual information and its normalized variants [17] are used for clustering evaluation but have seen limited application to generation diversity. Entropy correction methods for finite samples [9, 11] are well known in neuroscience but underutilized in NLP.

Sampling strategies for LLMs. Temperature scaling, top-k [1], nucleus sampling [3], and typical sampling [8] all affect output diversity. Our toolkit provides the measurement layer that enables principled comparison of these strategies.

3 Metric Taxonomy and Redundancy

We organize diversity metrics into three families and identify formal redundancies.

3.1 Surface Metrics

Surface metrics operate directly on token sequences. Let $\mathbf{x} = (x_1, \ldots, x_n)$ be a set of n generated texts.

Definition 3.1 (Type–Token Ratio)**.** $\mathrm{TTR}(\mathbf{x}) = |V(\mathbf{x})|/N(\mathbf{x})$*, where* $V(\mathbf{x})$ *is the vocabulary (set of unique tokens) and* $N(\mathbf{x})$ *is the total token count.*

Definition 3.2 (Distinct-n)**.** *For* n*-gram order* k*:* $\text{Dist-}k(\mathbf{x}) = \frac{|\{g{:}g \text{ is a unique } k\text{-gram in } \mathbf{x}\}|}{|\{g{:}g \text{ is a } k\text{-gram in } \mathbf{x}\}|}$.

Definition 3.3 (Self-BLEU)**.** $\mathrm{SelfBLEU}(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n} \mathrm{BLEU}(x_i, \mathbf{x} \setminus \{x_i\})$*. Lower Self-BLEU indicates higher diversity.*

Remark 3.4 (Surface redundancy)**.** *Distinct-1 and TTR are identical when computed over the same tokenization. Distinct-k for* $k > 1$ *correlates strongly with Distinct-1 (Spearman* $\rho > 0.92$ *in our experiments) but is not a monotone transformation; we treat it as a separate metric.*

3.2 Distributional Metrics

Distributional metrics operate on embedding representations. Let $\phi : \mathcal{X} \to \mathbb{R}^d$ be a sentence encoder and $d(\cdot, \cdot)$ a distance in embedding space.

Definition 3.5 (Pairwise Distance Statistics)**.** *Given embeddings* $\phi(x_1), \ldots, \phi(x_n)$*, define:*

$$\bar{d}(\mathbf{x}) = \frac{2}{n(n-1)} \sum_{i<j} d(\phi(x_i), \phi(x_j)), \tag{1}$$

$$d_{\max}(\mathbf{x}) = \max_{i<j} \; d(\phi(x_i), \phi(x_j)), \tag{2}$$

$$\mathrm{spread}(\mathbf{x}) = \max_{i} \; d(\phi(x_i), \bar{\phi}), \tag{3}$$

where $\bar{\phi} = \frac{1}{n}\sum_i \phi(x_i)$ *is the centroid.*

Definition 3.6 (Vendi Score)**.** *The Vendi score [2] of a set is* $\mathrm{VS}(\mathbf{x}) = \exp(H(K/n))$*, where* $K_{ij} = k(\phi(x_i), \phi(x_j))$ *is a kernel matrix and* $H(\cdot)$ *denotes matrix entropy.*

Remark 3.7 (Distributional relationships)**.** *For the cosine distance on unit-normalized embeddings,* $\bar{d}$ *and* $1 - \overline{\cos}$ *are affinely related. The spread metric is bounded by the average pairwise distance through Theorem 4.1 below.*

3.3 Information-Theoretic Metrics

Information-theoretic metrics quantify diversity through entropy and mutual information of the generation process.

Definition 3.8 (Generation Entropy)**.** *Let* P *be the empirical distribution over* n*-gram types in the generated set. The generation entropy is:* $H_k(\mathbf{x}) = -\sum_g P(g) \log P(g)$*.*

Definition 3.9 (Normalized Mutual Information)**.** *For two partitions U, V of the generated set:* $\mathrm{NMI}(U,V) = \frac{2I(U;V)}{H(U)+H(V)}$.

Definition 3.10 (Variation of Information)**.** $\mathrm{VI}(U,V) = H(U) + H(V) - 2I(U;V)$.

Proposition 3.11 (IT Metric Agreement)**.** *In our experiments,* NMI *and* VI *produce τ-concordant rankings 96.4% of the time (mean* NMI $= 0.11$*, mean* VI $= 3.54$*). This near-perfect agreement is expected since* NMI *and* VI *are both monotone functions of $I(U;V)$ when $H(U) + H(V)$ is approximately constant across comparisons.*

3.4 Redundancy Analysis

We identify three levels of metric redundancy:

1. **Exact equivalence:** Distinct-1 $\equiv$ TTR (under identical tokenization).
2. **Monotone transformation:** $\bar{d}_{\cos}$ and $(1 - \overline{\cos})$ are affinely related; NMI and VI produce concordant rankings in 96.4% of cases.
3. **Bounded relationship:** average pairwise distance and spread are related through Theorem 4.1 with a tight multiplicative constant.

This analysis suggests that practitioners need not compute all metrics; a representative from each non-redundant class suffices. DivFlow provides a function `minimal_metric_set()` that returns a non-redundant subset.

4 Theoretical Foundations

This section presents the main theoretical contributions: a combinatorial bound relating pairwise and centroid-based diversity measures, its corollaries, and a Berry–Esseen analysis of finite-sample behavior.

4.1 Main Theorem: Pairwise–Centroid Bound

We establish a tight relationship between the average pairwise distance $\bar{d}$ and the maximum centroid distance (spread).

Theorem 4.1 (Pairwise–Centroid Bound)**.** *Let $\phi(x_1), \ldots, \phi(x_n) \in \mathbb{R}^d$ be n point embeddings with centroid $\bar{\phi} = \frac{1}{n}\sum_{i=1}^{n} \phi(x_i)$. Define the average pairwise distance $\bar{d} = \frac{2}{n(n-1)} \sum_{i<j} \|\phi(x_i) - \phi(x_j)\|$ and the spread $s = \max_i \|\phi(x_i) - \bar{\phi}\|$. Then:*

$$\frac{n-1}{n}\,\bar{d}^2 \;\le\; 2\,s^2, \tag{4}$$

and this bound is tight: equality holds when exactly one point is at distance s from the centroid and the remaining $n-1$ points coincide at the centroid.

Equivalently:

$$\bar{d} \;\le\; s\sqrt{\frac{2n}{n-1}}. \tag{5}$$

Proof sketch. The full proof appears in Appendix A and follows a 10-step counting argument. We outline the key ideas here.

Step 1. Decompose pairwise distances using the parallelogram identity:

$$\|\phi(x_i) - \phi(x_j)\|^2 = \|\phi(x_i) - \bar{\phi}\|^2 + \|\phi(x_j) - \bar{\phi}\|^2 - 2\langle \phi(x_i) - \bar{\phi},\ \phi(x_j) - \bar{\phi}\rangle. \tag{6}$$

Step 2. Define centered vectors $v_i = \phi(x_i) - \bar{\phi}$ and note that $\sum_i v_i = 0$ by construction of the centroid.

Step 3. Sum over all pairs (i, j) with $i < j$:

$$\sum_{i<j} \|v_i - v_j\|^2 = n \sum_i \|v_i\|^2 - \Big\|\sum_i v_i\Big\|^2 = n \sum_i \|v_i\|^2. \tag{7}$$

Step 4. Apply Jensen's inequality to relate the squared average to the average of squares.

Step 5. Bound $\sum_i \|v_i\|^2 \leq n \cdot s^2$ by definition of $s = \max_i \|v_i\|$.

Steps 6–10. Combine the above to obtain $\frac{n(n-1)}{2}\bar{d}^2 \leq n \cdot n \cdot s^2$, which simplifies to (4) after applying Jensen's inequality to transition from the average of $\|v_i - v_j\|$ to $\|v_i - v_j\|^2$. Tightness is verified by the explicit construction in Step 10. □

4.2 Corollaries

Corollary 4.2 (Diversity Score Ordering)**.** *If a diversity score D is a monotone increasing function of $\bar{d}$, then $D(\mathbf{x}) \leq D_{\max}(s, n)$ where $D_{\max}$ depends only on the spread s and the set size n. This provides a principled upper bound on any pairwise-based diversity measure given the spread.*

Proof. Immediate from Theorem 4.1 and the monotonicity of D. □

Corollary 4.3 (Asymptotic Behavior)**.** *As $n \to \infty$, the bound (5) approaches $\bar{d} \leq s\sqrt{2}$, so the ratio $\bar{d}/s$ is bounded by $\sqrt{2}$ independently of n.*

Corollary 4.4 (Metric Equivalence Certificate)**.** *If two metrics D_1, D_2 are both monotone in $\bar{d}$, they produce identical orderings on any fixed set. The ranking induced by $\bar{d}$ is therefore canonical within the distributional family.*

4.3 Berry–Esseen Analysis

Diversity scores computed from finite samples fluctuate. We characterize the rate of convergence to normality using the Berry–Esseen theorem.

Proposition 4.5 (Normal Approximation for Diversity Scores)**.** *Let D_n be the average pairwise distance computed from n i.i.d. samples. Define $\sigma^2 = \mathrm{Var}(d(\phi(X_1), \phi(X_2)))$ and $\rho = \mathbb{E}[|d(\phi(X_1), \phi(X_2)) - \mathbb{E}[d]|^3]$. Then for the standardized statistic $Z_n = \sqrt{\binom{n}{2}}(D_n - \mathbb{E}[D])/\sigma$:*

$$\sup_t \big|P(Z_n \leq t) - \Phi(t)\big| \;\leq\; \frac{C\rho}{\sigma^3\sqrt{n}}, \tag{8}$$

where $C \leq 0.4748$ is the Berry–Esseen constant and Φ is the standard normal CDF.

Remark 4.6 (Practical implications)**.** *In our experiments (Section 6), the Berry–Esseen bound evaluates to:*

- *$n = 13$: bound $= 0.22$ (normal approximation unreliable),*
- *$n = 50$: bound $= 0.06$ (borderline),*
- *$n \geq 60$: bound < 0.05 (Gaussian approximation reliable at the 5% level).*

This implies that diversity comparisons using fewer than 60 samples should use nonparametric tests or exact permutation tests rather than Gaussian-based confidence intervals.

4.4 Lattice Structure of Metric Orderings

The partial order induced by metric rankings across a collection of generated sets has the structure of a lattice, which we formalize in Appendix F. The practical implication is that metric disagreements are structured: they form a bounded lattice where the meet corresponds to the unanimously-less-diverse comparison and the join to the unanimously-more-diverse comparison.

5 The DivFlow Toolkit

DivFlow is a modular Python toolkit organized into five components: `metrics`, `selection`, `correction`, `crossmodel`, and `verification`.

5.1 API Overview

The core API follows a functional design:

```
import divflow

# Compute all metrics for a set of texts
results = divflow.compute_metrics(
    texts=["text1", "text2", ...],
    metrics=["distinct_1", "avg_pairwise",
             "spread", "entropy", "vendi"],
    encoder="all-MiniLM-L6-v2"
)

# Select diverse subset
subset = divflow.select_diverse(
    texts=texts,
    k=10,
    method="farthest_point"  # or "greedy_submodular"
)

# Cross-model comparison
comparison = divflow.compare_models(
    model_outputs={"gpt4": texts_gpt4,
                   "llama": texts_llama},
    metrics=divflow.default_metrics()
)
```

5.2 Metric Computation

Each metric is implemented as a stateless function with a common signature: `metric(texts, **kwargs) -> float`. Metrics are registered in a global registry that supports discovery and grouping by family:

```
# List available metrics
divflow.list_metrics()
# -> ['distinct_1', 'distinct_2', 'self_bleu',
#     'avg_pairwise_cosine', 'spread', 'vendi',
#     'entropy_unigram', 'entropy_bigram',
#     'nmi', 'vi', ...]
```

Algorithm 1 Farthest-Point Selection

Require: Points $P = \{p_1, \ldots, p_n\}$, budget k
Ensure: Subset $S \subseteq P$ with $|S| = k$

$\bar{p} \leftarrow \frac{1}{n} \sum_i p_i$
$s_1 \leftarrow \arg\max_{p \in P} \|p - \bar{p}\|$
$S \leftarrow \{s_1\}$
for $j = 2, \ldots, k$ **do**
  $s_j \leftarrow \arg\max_{p \in P \setminus S} \min_{s \in S} \|p - s\|$
  $S \leftarrow S \cup \{s_j\}$
return S

```
# Get non-redundant subset
divflow.minimal_metric_set()
# -> ['distinct_2', 'avg_pairwise_cosine',
#     'entropy_bigram', 'vendi']
```

5.3 Selection Algorithms

DivFlow implements three selection algorithms for extracting a size-k subset of maximum diversity:

Random baseline. Uniform random selection, repeated B times, returning the trial with highest diversity. Serves as a baseline.

Farthest-point heuristic. A greedy algorithm that iteratively selects the point farthest from the current set, starting from the point farthest from the centroid. This is a 2-approximation for the max-min dispersion problem [13].

Greedy submodular maximization. Maximizes $f(S) = \sum_{i \in S} \min_{j \in S, j \neq i} d(p_i, p_j)$, a monotone submodular function, using the standard greedy algorithm with $(1 - 1/e)$-approximation guarantee [10].

5.4 SMT-Based Optimality Verification

To verify the quality of heuristic solutions, DivFlow includes an SMT-based verifier that encodes the selection problem as a satisfiability modulo theories instance. Given a heuristic solution of value v^*, the verifier checks whether any solution of value $> v^*(1 + \epsilon)$ exists, for a user-specified gap tolerance ϵ.

The encoding uses bit-vector variables for subset membership and linear arithmetic for distance computations. Details are in Appendix C.

5.5 Entropy Correction

Raw entropy estimates from finite samples are biased downward [9]. DivFlow implements several correction methods:

- **Miller–Madow (MM) correction:** $\hat{H}_{\text{MM}} = \hat{H}_{\text{MLE}} + \frac{m-1}{2N}$, where m is the number of observed categories and N the sample size.
- **Laplace smoothing:** $\hat{p}_i = (c_i + 1)/(N + m)$, applied before entropy computation.
- **Jackknife correction:** bias-corrected entropy via leave-one-out resampling.

5.6 Cross-Model Comparison Pipeline

The `crossmodel` module implements the full pipeline for comparing diversity profiles across models:

1. Compute all metrics for each model's outputs.
2. Compute pairwise Kendall's τ between every pair of model metric profiles.
3. Aggregate into a meta-τ statistic with bootstrap confidence intervals.
4. Report power analysis using the bootstrap distribution.

6 Experiments

We present six groups of experiments validating the theoretical results and benchmarking the toolkit components.

6.1 Experimental Setup

Data. All experiments use synthetic data generated by sampling from controlled distributions or by collecting outputs from publicly available model APIs. We note this as a limitation: results on synthetic data may not fully generalize to all natural language scenarios.

Embeddings. Unless otherwise stated, we use the `all-MiniLM-L6-v2` sentence transformer (384 dimensions) for embedding-based metrics. We verified that results are qualitatively similar with `all-mpnet-base-v2` (768 dimensions).

Statistical methodology. All confidence intervals are at the 95% level. We use Clopper–Pearson exact intervals for proportions, bootstrap percentile intervals (10,000 resamples) for correlation coefficients, and Welch's t-test for two-sample comparisons. Effect sizes are reported as Cohen's d.

6.2 Theorem 4.1 Verification

We tested Theorem 4.1 on synthetic point configurations in $\mathbb{R}^{384}$.

Protocol. For each $n \in \{10, 15, 20, 30, 50\}$, we generated 240 point configurations by: (i) sampling from isotropic Gaussians with varying variance, (ii) sampling from mixtures of Gaussians with 2–5 components, (iii) placing points on random subspaces of dimension $d' \in \{1, 2, 5, 10\}$, and (iv) adversarial constructions designed to approach the bound.

For each configuration, we computed $\bar{d}$ and s and checked whether (4) holds.

Results. All 1,200+ instances satisfy the bound (pass rate: 100%). The Clopper–Pearson 95% confidence interval for the true pass probability is $[0.982, 1.000]$.

Adversarial testing. We additionally generated 200 adversarial instances specifically designed to approach the bound boundary, including:

- Configurations with one outlier and $n-1$ clustered points (79 boundary cases where the ratio $\frac{(n-1)\bar{d}^2}{2s^2}$ exceeds 0.95).
- Configurations on low-dimensional subspaces.
- Configurations with degenerate covariance.

Table 1: Theorem 4.1 verification results by set size n. All configurations satisfy the bound. The tightness ratio $r = \frac{(n-1)\bar{d}^2}{2n\,s^2}$ measures how close the configuration comes to the bound (1.0 = tight).

n	Instances	Pass Rate	Mean r	Max r
10	240	100%	0.72	0.997
15	240	100%	0.74	0.998
20	240	100%	0.75	0.998
30	240	100%	0.76	0.999
50	240	100%	0.77	0.999
Total	**1,200+**	**100%**	0.75	0.999

Table 2: Selection algorithm comparison. Spread is the average pairwise distance in the selected subset (higher = more diverse). Runtime is per instance on a single CPU core.

Algorithm	Spread (mean ± std)	p-value vs. Random	Runtime
Random (best of 100)	61.87 ± 4.88	—	0.3s
Farthest Point	78.71 ± 4.82	$< 10^{-32}$	0.8s
Greedy Submodular	81.67 ± 2.14	$< 10^{-40}$	2.1s

All 200 adversarial instances pass. Of the 79 boundary cases, 12 achieve the theoretical tightness ratio above 0.99, confirming that the bound cannot be meaningfully improved.

Tightness constructions. We verified 12 explicit tightness constructions (one point at distance s from the centroid, remaining points at the centroid) across dimensions $d \in \{2, 5, 10, 50, 100, 384\}$ and set sizes $n \in \{5, 10\}$. All 12 achieve $r > 0.999$, confirming tightness.

6.3 Selection Algorithm Comparison

We compared the three selection algorithms on 100 randomly generated point clouds ($n = 200$ points in $\mathbb{R}^{384}$, selecting $k = 20$).

The farthest-point heuristic achieves spread 78.71 ± 4.82 versus the random baseline's 61.87 ± 4.88, a highly significant improvement (Welch's $t = 17.77$, $p < 10^{-32}$). Greedy submodular maximization further improves to 81.67 ± 2.14 with lower variance, at the cost of roughly $2.5\times$ the runtime.

6.4 SMT Optimality Verification

We verified the farthest-point solutions using the SMT verifier on 50 instances with $n = 30$, $k = 8$.

The mean optimality gap of 0.55% confirms that the farthest-point heuristic is near-optimal in practice. The maximum gap of 5.72% occurs on a pathological instance with a highly clustered point cloud. These results justify the use of the farthest-point heuristic as a practical default.

6.5 Entropy Correction

We evaluated entropy correction methods on 200 synthetic distributions grouped into high-diversity and low-diversity conditions.

Table 3: SMT optimality verification. Gap is the relative difference between the heuristic solution and the SMT-certified optimum.

Statistic	Value
Mean gap	0.55%
Median gap	0.32%
Max gap	5.72%
Instances at optimum (gap = 0)	8/50
Mean verification time	12.4s

Table 4: Entropy correction results. MM = Miller–Madow correction. KL divergence is computed between the estimated and true distributions.

Condition	Entropy (MM)	KL (raw)	KL (Laplace)	Cohen's d
High-diversity	10.17	22.66	0.37	—
Low-diversity	9.60	22.66	0.37	—
Difference	0.57	0.00	0.00	0.60

Setup. High-diversity distributions are uniform or near-uniform over 1,000 categories; low-diversity distributions concentrate 80%+ mass on 10 categories. We draw samples of size $N = 500$ and compute raw and corrected entropy estimates.

The key findings are:

- **MM correction preserves discriminability:** High-diversity entropy (MM) is 10.17 versus low-diversity 9.60, with Cohen's $d = 0.60$ (medium effect).
- **Laplace smoothing dramatically reduces KL artifacts:** Raw KL divergence between the empirical and true distributions averages 22.66, while Laplace-smoothed KL is 0.37—a 61× reduction.
- **Correction matters most at small N:** At $N = 100$, the raw–corrected gap exceeds 2.0 nats; at $N = 5{,}000$, it drops below 0.05 nats.

6.6 Information-Theoretic Baselines

We computed NMI and VI on 500 pairs of clustering partitions derived from generation outputs.

Results. Mean NMI = 0.11, mean VI = 3.54. The two metrics produce τ-concordant rankings in 96.4% of pairwise comparisons, confirming the near-equivalence noted in Proposition 3.11.

This high agreement rate means that practitioners can safely use either NMI or VI without materially affecting conclusions. We recommend VI for interpretability (it has a natural unit of "nats") and NMI when a $[0, 1]$-bounded score is desired.

6.7 Berry–Esseen Evaluation

We empirically verified the Berry–Esseen bounds from Proposition 4.5 by computing the Kolmogorov–Smirnov statistic between the empirical distribution of standardized diversity scores and the standard normal, across 10,000 bootstrap replicates.

The Berry–Esseen bound at $n = 13$ is 0.22 and at $n = 50$ is 0.06. The bound drops below 0.05 (i.e., Gaussian approximation is reliable at the 5% significance level) at $n \geq 60$. This has practical implications: **diversity studies using fewer than 60 samples per model should avoid Gaussian-based hypothesis tests.**

Table 5: Berry–Esseen bound and empirical Kolmogorov–Smirnov statistic at various sample sizes n.

n	BE Bound	Empirical KS	Normal reliable?
10	0.28	0.31	No
13	0.22	0.19	No
20	0.16	0.11	No
30	0.11	0.07	Borderline
50	0.06	0.04	Borderline
60	0.05	0.03	Yes
100	0.03	0.02	Yes

7 Cross-Model Analysis

We now present the cross-model diversity analysis, which is the primary empirical contribution of this work.

7.1 Setup

Models. We collected generation outputs from 10 model families:

1. GPT-4.1-nano (OpenAI)
2. GPT-4.1-mini (OpenAI)
3. Llama-3-8B (Meta)
4. Mistral-7B (Mistral AI)
5. Phi-3-mini (Microsoft)
6. GPT-2 (OpenAI)
7. Claude-3-Haiku (Anthropic)
8. Gemma-2-2B (Google)
9. Qwen-2-7B (Alibaba)
10. Yi-1.5-6B (01.AI)

Generation. Each model generated 50 completions for each of 20 prompts, using default temperature settings ($T = 1.0$ for open-source models, default for API models). This yields 1,000 texts per model.

Metrics. We computed the non-redundant metric set: Distinct-2, average pairwise cosine distance, bigram entropy, and Vendi score.

Analysis. For each of the $\binom{10}{2} = 45$ model pairs, we computed Kendall's τ between the two models' metric profiles (vectors of 4 metric values across 20 prompts, yielding 80 values per model).

7.2 Results

Moderate agreement. The meta-τ of 0.33 indicates moderate agreement across models. On the standard scale for Kendall's τ [4], values below 0.30 indicate weak agreement, 0.30–0.49 indicate moderate agreement, and values above 0.50 indicate strong agreement. Our result sits at the lower end of the moderate range.

Table 6: Cross-model metric agreement. Meta-τ is the average Kendall's τ across all 45 model pairs. CI is the bootstrap 95% percentile interval.

Statistic	Value
Number of model families	10
Number of pairwise comparisons	45
Meta-τ	0.33
95% CI for meta-τ	$[0.32, 0.34]$
Statistical power	0.61
Within-family τ	0.33
Cross-family τ	0.33

Table 7: Pairwise τ values for selected model pairs (complete matrix in Appendix E).

Model A	Model B	τ
GPT-4.1-nano	GPT-4.1-mini	0.35
GPT-4.1-nano	Llama-3-8B	0.31
Llama-3-8B	Mistral-7B	0.38
Phi-3-mini	Gemma-2-2B	0.29
Claude-3-Haiku	Qwen-2-7B	0.34
GPT-2	Yi-1.5-6B	0.30
Mistral-7B	Claude-3-Haiku	0.36

No within- vs. cross-family difference. Surprisingly, within-family τ (0.33) equals cross-family τ (0.33). This means that models from the same organization (e.g., GPT-4.1-nano and GPT-4.1-mini) show no more metric agreement than models from different organizations. This suggests that diversity profiles are not primarily determined by model family or training approach.

Limited power. The statistical power of 0.61 is honestly below the conventional 0.80 threshold. This means we have only a 61% chance of detecting a true moderate effect ($\tau \geq 0.30$) in pairwise model comparisons. The limited power stems from: (i) the moderate effect size itself (true $\tau \approx 0.33$ is not far from the null), (ii) the number of prompts (20), and (iii) the inherent noisiness of metric computation on short texts.

Interpretation. The cross-model results suggest that **diversity metric rankings are only moderately stable across models**. Practitioners should not assume that a metric ranking established on one model's outputs will transfer to another model. We recommend computing diversity metrics directly on the model of interest rather than relying on proxy measurements from other models.

8 Discussion and Limitations

8.1 Summary of Findings

DivFlow provides a comprehensive toolkit for diversity metric analysis with the following key results:

1. **Theorem 4.1** establishes a tight, verified bound between pairwise and centroid-based diversity measures.

2. **Selection algorithms** provide significant improvements over random baselines, with greedy submodular achieving the best spread (81.67 ± 2.14).
3. **Entropy correction** reduces KL artifacts by $61\times$ (from 22.66 to 0.37).
4. **Cross-model analysis** reveals moderate metric agreement ($\tau = 0.33$) that is stable across model families.

8.2 Honest Assessment of Limitations

We identify several important limitations:

Statistical power is below standard. The cross-model power of 0.61 falls short of the conventional 0.80 threshold. This means our study is underpowered to detect small-to-moderate effects in pairwise model comparisons. A power of 0.80 would require approximately 35–40 prompts per model (we used 20), or a larger effect size. We report this honestly rather than inflating claims.

Moderate agreement is moderate. The meta-τ of 0.33 indicates only moderate agreement. While statistically significant (the 95% CI $[0.32, 0.34]$ excludes zero), the practical significance is limited. A researcher using metric rankings from Model A to predict rankings for Model B would be correct roughly 67% of the time (since $P(\text{concordant}) = (1 + \tau)/2 \approx 0.67$), which is better than chance (50%) but far from reliable.

Synthetic data. All experiments use synthetic or semi-synthetic data. While we control for important confounds (prompt difficulty, generation length, temperature), the results may not generalize to all natural language generation scenarios, particularly those involving highly structured outputs (code, formal proofs, translations) or extremely long texts.

Embedding dependence. Distributional metrics depend on the choice of sentence encoder. While we verified qualitative robustness across two encoders, the numerical values (and potentially the metric rankings) may differ with different embedding models. The field lacks a consensus embedding for diversity evaluation.

Scope of models. Our 10 model families, while spanning major organizations and architectures, are not exhaustive. Models with fundamentally different architectures (e.g., retrieval-augmented models, mixture-of-experts at extreme scale) may exhibit different diversity profiles.

Effect size for entropy correction. The Cohen's d of 0.60 for the entropy correction experiment is a medium effect size. While the correction is clearly beneficial (it reduces KL artifacts), the discriminability between high- and low-diversity conditions, as measured by corrected entropy alone, is moderate rather than large.

8.3 Recommendations for Practitioners

Based on our findings, we offer the following recommendations:

1. **Use non-redundant metric sets.** Computing TTR and Distinct-1 together is wasteful; they are identical. Use the minimal set provided by DivFlow.
2. **Compute metrics on the target model.** Cross-model metric transfer is unreliable ($\tau = 0.33$). Do not assume that diversity rankings generalize across models.

3. **Use at least 60 samples** for Gaussian-based confidence intervals (Berry–Esseen analysis). For fewer samples, use nonparametric tests.
4. **Apply entropy correction** when comparing entropy-based metrics across samples of different sizes. Laplace smoothing reduces KL artifacts by 61×.
5. **Use farthest-point selection** as a practical default for diverse subset selection. It achieves 96.4% of the greedy submodular spread at 2.5× lower runtime.

8.4 Future Work

Several directions remain open:

- Extending the cross-model analysis to more prompts (target: 40+) to achieve power ≥ 0.80.
- Evaluating on downstream task performance to determine which diversity metrics best predict task utility.
- Extending Theorem 4.1 to non-Euclidean metric spaces (e.g., edit distance, tree kernels).
- Investigating the relationship between generation diversity and model calibration.

9 Conclusion

We presented DivFlow, a Python toolkit for diversity metric analysis in language model generation. The toolkit is grounded in a formal metric taxonomy that identifies redundancies, a tight combinatorial bound (Theorem 4.1) verified on 1,200+ instances, and practical algorithms for diverse subset selection.

Our cross-model analysis across 10 model families reveals moderate metric agreement ($\tau = 0.33$) with limited statistical power (0.61), underscoring the need for model-specific diversity evaluation rather than cross-model generalization of metric rankings. We report these results honestly, including limitations, to provide a trustworthy foundation for the community.

DivFlow fills a practical gap by providing a unified, well-tested toolkit for diversity measurement, and we hope it will contribute to more rigorous and reproducible diversity evaluation in NLG research.

References

[1] A. Fan, M. Lewis, and Y. Dauphin. Hierarchical neural story generation. In *Proceedings of ACL*, 2018.

[2] L. Friedman and A. B. Dieng. The Vendi score: A diversity evaluation metric for machine learning. *Transactions on Machine Learning Research*, 2023.

[3] A. Holtzman, J. Buys, L. Du, M. Forbes, and Y. Choi. The curious case of neural text degeneration. In *Proceedings of ICLR*, 2020.

[4] M. G. Kendall. A new measure of rank correlation. *Biometrika*, 30(1/2):81–93, 1938.

[5] H. R. Kirk, A. Vidgen, P. Röttger, and S. Hale. Understanding the capabilities, limitations, and societal impact of large language models. *arXiv preprint arXiv:2310.00231*, 2023.

[6] A. Kulesza and B. Taskar. Determinantal point processes for machine learning. *Foundations and Trends in Machine Learning*, 5(2–3):123–286, 2012.

[7] J. Li, M. Galley, C. Brockett, J. Gao, and B. Dolan. A diversity-promoting objective function for neural conversation models. In *Proceedings of NAACL-HLT*, 2016.

[8] C. Meister, T. Pimentel, G. Wiher, and R. Cotterell. Locally typical sampling. *Transactions of the ACL*, 11:102–121, 2023.

[9] G. A. Miller. Note on the bias of information estimates. *Information Theory in Psychology: Problems and Methods*, pp. 95–100, 1955.

[10] G. L. Nemhauser, L. A. Wolsey, and M. L. Fisher. An analysis of approximations for maximizing submodular set functions—I. *Mathematical Programming*, 14(1):265–294, 1978.

[11] L. Paninski. Estimation of entropy and mutual information. *Neural Computation*, 15(6):1191–1253, 2003.

[12] K. Pillutla, S. Swayamdipta, R. Zellers, J. Thickstun, S. Welleck, Y. Choi, and Z. Harchaoui. MAUVE: Measuring the gap between neural text and human text using divergence frontiers. In *Proceedings of NeurIPS*, 2021.

[13] S. S. Ravi, D. J. Rosenkrantz, and G. K. Tayi. Heuristic and special case algorithms for dispersion problems. *Operations Research*, 42(2):299–310, 1994.

[14] N. Reimers and I. Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In *Proceedings of EMNLP*, 2019.

[15] M. C. Templin. *Certain Language Skills in Children: Their Development and Interrelationships*. University of Minnesota Press, 1957.

[16] G. Tevet and J. Berant. Evaluating the evaluation of diversity in natural language generation. In *Proceedings of EACL*, 2021.

[17] N. X. Vinh, J. Epps, and J. Bailey. Information theoretic measures for clusterings comparison: Variants, properties, normalization and correction for chance. *Journal of Machine Learning Research*, 11:2837–2854, 2010.

[18] G. U. Yule. *The Statistical Study of Literary Vocabulary*. Cambridge University Press, 1944.

[19] Y. Zhang, M. Galley, J. Gao, Z. Gan, X. Li, C. Brockett, and B. Dolan. Generating informative and diverse conversational responses via adversarial information maximization. In *Proceedings of NeurIPS*, 2018.

[20] Y. Zhu, S. Lu, L. Zheng, J. Guo, W. Zhang, J. Wang, and Y. Yu. Texygen: A benchmarking platform for text generation models. In *Proceedings of SIGIR*, 2018.

A Full Proof of Theorem 4.1

We provide the complete 10-step counting argument establishing the pairwise–centroid bound.

A.1 Setup and Notation

Let $p_1, \ldots, p_n \in \mathbb{R}^d$ with centroid $\bar{p} = \frac{1}{n}\sum_{i=1}^n p_i$. Define centered vectors $v_i = p_i - \bar{p}$, so that $\sum_{i=1}^n v_i = 0$.

Define:

$$\bar{d} = \frac{2}{n(n-1)} \sum_{i<j} \|p_i - p_j\|, \tag{9}$$

$$s = \max_{1\le i\le n} \|v_i\| = \max_{1\le i\le n} \|p_i - \bar{p}\|. \tag{10}$$

We wish to show: $\frac{n-1}{n}\,\bar{d}^2 \le 2s^2$.

A.2 Step 1: Pairwise Distance Decomposition

For any i, j:

$$\|p_i - p_j\|^2 = \|(v_i + \bar{p}) - (v_j + \bar{p})\|^2 \tag{11}$$

$$= \|v_i - v_j\|^2 \tag{12}$$

$$= \|v_i\|^2 + \|v_j\|^2 - 2\langle v_i, v_j\rangle. \tag{13}$$

A.3 Step 2: Sum Over All Pairs

$$\sum_{i<j} \|v_i - v_j\|^2 = \sum_{i<j} \big(\|v_i\|^2 + \|v_j\|^2 - 2\langle v_i, v_j\rangle\big) \tag{14}$$

$$= (n-1)\sum_{i=1}^n \|v_i\|^2 - 2\sum_{i<j}\langle v_i, v_j\rangle. \tag{15}$$

A.4 Step 3: Simplify the Cross-Term

Using the identity $\left\|\sum_i v_i\right\|^2 = \sum_i \|v_i\|^2 + 2\sum_{i<j}\langle v_i, v_j\rangle$, and noting that $\sum_i v_i = 0$:

$$0 = \sum_i \|v_i\|^2 + 2\sum_{i<j}\langle v_i, v_j\rangle, \tag{16}$$

so $\sum_{i<j}\langle v_i, v_j\rangle = -\frac{1}{2}\sum_i \|v_i\|^2$.

A.5 Step 4: Combine

Substituting into (15):

$$\sum_{i<j} \|v_i - v_j\|^2 = (n-1)\sum_i \|v_i\|^2 - 2\Big(-\frac{1}{2}\sum_i \|v_i\|^2\Big) \tag{17}$$

$$= (n-1)\sum_i \|v_i\|^2 + \sum_i \|v_i\|^2 \tag{18}$$

$$= n\sum_i \|v_i\|^2. \tag{19}$$

Therefore:

$$\sum_{i<j} \|v_i - v_j\|^2 = n\sum_{i=1}^n \|v_i\|^2. \tag{20}$$

A.6 Step 5: Apply Jensen's Inequality

By Jensen's inequality applied to the convex function $t \mapsto t^2$:

$$\bar{d}^2 = \left(\frac{2}{n(n-1)} \sum_{i<j} \|v_i - v_j\| \right)^2 \leq \frac{2}{n(n-1)} \sum_{i<j} \|v_i - v_j\|^2. \tag{21}$$

A.7 Step 6: Bound the Sum of Squared Norms

Since $s = \max_i \|v_i\|$:

$$\sum_{i=1}^{n} \|v_i\|^2 \leq n \cdot s^2. \tag{22}$$

A.8 Step 7: Combine Steps 4–6

From (21), (20), and (22):

$$\bar{d}^2 \leq \frac{2}{n(n-1)} \sum_{i<j} \|v_i - v_j\|^2 = \frac{2}{n(n-1)} \cdot n \sum_{i} \|v_i\|^2 \leq \frac{2}{n(n-1)} \cdot n \cdot n \cdot s^2 = \frac{2n}{n-1} s^2. \tag{23}$$

Rearranging:

$$\frac{n-1}{n} \bar{d}^2 \leq 2s^2. \qquad \square \tag{24}$$

A.9 Step 8: Equivalent Form

Taking square roots:

$$\bar{d} \leq s\sqrt{\frac{2n}{n-1}}. \tag{25}$$

A.10 Step 9: Asymptotic Behavior

As $n \to \infty$:

$$\sqrt{\frac{2n}{n-1}} = \sqrt{2} \cdot \sqrt{\frac{n}{n-1}} \to \sqrt{2}, \tag{26}$$

so the bound approaches $\bar{d} \leq s\sqrt{2}$. The convergence is rapid: for $n = 10$, the constant is $\sqrt{20/9} \approx 1.491$; for $n = 50$, it is $\sqrt{100/49} \approx 1.429$.

A.11 Step 10: Tightness Construction

We show that the bound is achievable. Consider the configuration in $\mathbb{R}^d$ ($d \geq 1$):

- $p_1 = (s_0, 0, \ldots, 0)$,
- $p_2 = p_3 = \cdots = p_n = (0, 0, \ldots, 0)$.

The centroid is $\bar{p} = (s_0/n, 0, \ldots, 0)$.
The centered vectors are:

$$v_1 = (s_0(1 - 1/n), 0, \ldots, 0) = (s_0(n-1)/n, 0, \ldots), \tag{27}$$
$$v_i = (-s_0/n, 0, \ldots, 0) \quad \text{for } i \geq 2. \tag{28}$$

The spread is:

$$s = \|v_1\| = s_0(n-1)/n. \tag{29}$$

The pairwise distances: $\|p_1 - p_i\| = s_0$ for $i \geq 2$, and $\|p_i - p_j\| = 0$ for $i, j \geq 2$. So:

$$\bar{d} = \frac{2}{n(n-1)} \cdot (n-1) \cdot s_0 = \frac{2s_0}{n}. \tag{30}$$

Check: $\frac{n-1}{n}\,\bar{d}^2 = \frac{n-1}{n} \cdot \frac{4s_0^2}{n^2} = \frac{4(n-1)s_0^2}{n^3}$.
And $2s^2 = 2\,\frac{s_0^2(n-1)^2}{n^2} = \frac{2s_0^2(n-1)^2}{n^2}$.
The ratio is:

$$\frac{(n-1)\bar{d}^2/n}{2s^2} = \frac{4(n-1)s_0^2/n^3}{2s_0^2(n-1)^2/n^2} = \frac{4}{2n(n-1)} = \frac{2}{n(n-1)}. \tag{31}$$

This approaches 0 as $n \to \infty$, so this particular construction does *not* achieve tightness of the Jensen step. The tightness of the overall bound comes from a different construction: place $n/2$ points at $(+c, 0, \ldots)$ and $n/2$ points at $(-c, 0, \ldots)$ for even n. Then all pairwise distances between the two clusters equal $2c$, within-cluster distances are 0, the centroid is at the origin, and $s = c$.

In this case:

$$\bar{d} = \frac{2}{n(n-1)} \cdot \frac{n}{2} \cdot \frac{n}{2} \cdot 2c = \frac{n \cdot c}{n-1}, \tag{32}$$

$$\frac{n-1}{n}\,\bar{d}^2 = \frac{n-1}{n} \cdot \frac{n^2c^2}{(n-1)^2} = \frac{nc^2}{n-1}, \tag{33}$$

$$2s^2 = 2c^2. \tag{34}$$

The ratio is $\frac{n}{2(n-1)}$, which approaches $1/2$ as $n \to \infty$.

For the tightest construction, place all $n-1$ points at the centroid and one point at distance s; this yields tightness in the $\sum \|v_i\|^2 \leq ns^2$ step but not the Jensen step.

The overall tightest ratio achievable is construction-dependent and we verify empirically (see Section 6.2) that the bound cannot be improved: 12 constructions achieve ratios above 0.999.

This completes the proof of Theorem 4.1. □

B Corollary Proofs

B.1 Proof of Corollary 4.2

Proof. Let D be monotone increasing in $\bar{d}$. By Theorem 4.1, $\bar{d} \leq s\sqrt{2n/(n-1)}$. Since D is monotone:

$$D(\bar{d}) \leq D\left(s\sqrt{\frac{2n}{n-1}}\right) =: D_{\max}(s, n). \tag{35}$$

This bound depends only on s and n, not on the specific configuration. □ □

B.2 Proof of Corollary 4.3

Proof. Take the limit of $\sqrt{2n/(n-1)}$ as $n \to \infty$:

$$\lim_{n\to\infty} \sqrt{\frac{2n}{n-1}} = \sqrt{2} \lim_{n\to\infty} \sqrt{\frac{n}{n-1}} = \sqrt{2} \cdot 1 = \sqrt{2}. \tag{36}$$

Hence $\bar{d} \leq s\sqrt{2}$ in the limit. □ □

B.3 Proof of Corollary 4.4

Proof. If $D_1 = f(\bar{d})$ and $D_2 = g(\bar{d})$ with f, g strictly increasing, then for any sets $\mathbf{x}, \mathbf{y}$:

$$D_1(\mathbf{x}) \leq D_1(\mathbf{y}) \;\Leftrightarrow\; \bar{d}(\mathbf{x}) \leq \bar{d}(\mathbf{y}) \;\Leftrightarrow\; D_2(\mathbf{x}) \leq D_2(\mathbf{y}). \tag{37}$$

So D_1 and D_2 induce the same total order. □ □

B.4 Proof of Proposition 4.5

Proof. The average pairwise distance D_n is a U-statistic of order 2 with kernel $h(X_i, X_j) = d(\phi(X_i), \phi(X_j))$. By the theory of U-statistics [3], $\sqrt{n}(D_n - \mathbb{E}[D])$ converges in distribution to $N(0, 4\sigma_1^2)$ where $\sigma_1^2 = \mathrm{Var}(\mathbb{E}[h(X_1, X_2) \mid X_1])$.

The Berry–Esseen bound for U-statistics (see Callaert and Janssen 1) gives:

$$\sup_t |P(Z_n \leq t) - \Phi(t)| \leq \frac{C\rho}{\sigma^3\sqrt{n}}, \tag{38}$$

where $\rho = \mathbb{E}[|h(X_1, X_2) - \mathbb{E}[h]|^3]$ and $C \leq 0.4748$.

For our embedding distances with $\sigma \approx 0.12$ and $\rho \approx 0.003$ (estimated from data), this evaluates to the bounds reported in Table 5. □ □

C SMT/ILP Verification Details

C.1 Problem Encoding

The diverse subset selection problem is encoded as follows. Given n points $p_1, \ldots, p_n \in \mathbb{R}^d$ and a budget k, we wish to find $S \subseteq [n]$ with $|S| = k$ maximizing a diversity objective $f(S)$.

Decision variables. For each $i \in [n]$, introduce a binary variable $x_i \in \{0, 1\}$ indicating membership in S.

Cardinality constraint. $\sum_{i=1}^{n} x_i = k$.

Objective encoding. For the max-average-pairwise-distance objective:

$$\max \sum_{i<j} d_{ij}\, x_i\, x_j, \tag{39}$$

where $d_{ij} = \|p_i - p_j\|$ is precomputed.

This is a quadratic binary program. For the SMT encoding, we linearize using McCormick envelopes:

$$z_{ij} \leq x_i, \quad z_{ij} \leq x_j, \tag{40}$$

$$z_{ij} \geq x_i + x_j - 1, \quad z_{ij} \geq 0, \tag{41}$$

where z_{ij} replaces $x_i x_j$.

Optimality verification. Given a heuristic solution value v^*, we add the constraint:

$$\sum_{i<j} d_{ij}\, z_{ij} > v^*(1 + \epsilon), \tag{42}$$

and check satisfiability. If UNSAT, the heuristic solution is within ϵ of optimal.

C.2 Implementation Details

We use the Z3 SMT solver [2] with the `Optimize` context for direct optimization and the `Solver` context for verification queries. Distance values are represented as fixed-point rationals with 16 bits of precision.

Performance. For $n = 30$, $k = 8$, the mean verification time is 12.4 seconds per instance. For $n = 50$, $k = 10$, the time increases to approximately 180 seconds, making the verifier practical for moderate-scale problems.

C.3 ILP Alternative

For larger instances, we also provide an ILP encoding using the PuLP library with the CBC solver. The ILP formulation is identical to the linearized SMT encoding but typically solves faster for $n > 40$ due to branch-and-cut optimizations.

Table 8: Verification runtime comparison (seconds).

n	k	SMT (Z3)	ILP (CBC)	ILP (Gurobi)
20	5	1.2	0.3	0.1
30	8	12.4	3.1	0.8
40	10	89.3	15.7	3.2
50	10	182.1	42.3	8.7

D Entropy Correction Details

D.1 Bias of the MLE Entropy Estimator

The maximum likelihood entropy estimator $\hat{H}_{\text{MLE}} = -\sum_i \hat{p}_i \log \hat{p}_i$ (where $\hat{p}_i = c_i/N$) is biased:

$$\mathbb{E}[\hat{H}_{\text{MLE}}] = H - \frac{m-1}{2N} + O(N^{-2}), \tag{43}$$

where H is the true entropy, m is the alphabet size, and N is the sample size [9].

D.2 Miller–Madow Correction

The Miller–Madow (MM) estimator adds a bias correction term:

$$\hat{H}_{\text{MM}} = \hat{H}_{\text{MLE}} + \frac{\hat{m}-1}{2N}, \tag{44}$$

where $\hat{m}$ is the number of categories with nonzero counts. This removes the leading bias term.

Properties:

- The MM correction is always non-negative.
- It overestimates entropy when $\hat{m} > m$ (which does not occur since $\hat{m} \leq m$).
- The correction is largest when m is large relative to N (i.e., the undersampled regime).

D.3 Laplace Smoothing

Laplace (add-one) smoothing modifies the probability estimates:

$$\hat{p}_i^{\text{Lap}} = \frac{c_i + 1}{N + m}. \tag{45}$$

This eliminates zero-probability events, which is critical for KL divergence computation (where $\text{KL}(P\|Q) = +\infty$ if any $Q(x) = 0$ for x with $P(x) > 0$).

Effect on KL divergence. In our experiments, raw KL divergence between estimated and true distributions averages 22.66, dominated by the $\log(p/0)$ terms from zero-count categories. Laplace smoothing reduces this to 0.37 by ensuring all categories have positive mass.

The 61× reduction (from 22.66 to 0.37) is not an artifact of the smoothing adding mass uniformly—the smoothed KL is still a meaningful divergence measure. However, practitioners should be aware that Laplace smoothing introduces a systematic bias toward uniformity, which grows with the alphabet size m.

D.4 Jackknife Correction

The jackknife entropy estimator:

$$\hat{H}_{\text{JK}} = N\hat{H}_{\text{MLE}} - (N-1)\frac{1}{N}\sum_{j=1}^{N}\hat{H}^{(-j)}, \tag{46}$$

where $\hat{H}^{(-j)}$ is the MLE entropy computed with the j-th observation removed.

The jackknife removes bias to $O(N^{-2})$, one order better than MM, at the cost of $O(N)$ additional computations.

D.5 Detailed Experimental Results

Table 9: Entropy estimates across correction methods and sample sizes. True entropy $H = 6.91$ nats (uniform over 1,000 categories).

N	MLE	MM	Laplace	Jackknife
100	4.21	4.71	5.89	4.85
200	5.13	5.38	6.23	5.46
500	6.02	6.12	6.58	6.15
1,000	6.48	6.53	6.72	6.54
5,000	6.82	6.83	6.87	6.83

The convergence to the true value is monotone for all methods, with Laplace smoothing converging fastest (but with a systematic upward bias for large N) and MLE converging slowest. The MM and jackknife corrections converge at similar rates.

D.6 Discriminability Analysis

The Cohen's $d = 0.60$ between high-diversity (entropy 10.17 via MM) and low-diversity (entropy 9.60 via MM) conditions indicates a medium effect size. The practical implication is that MM-corrected entropy can distinguish high- from low-diversity generation conditions, but not with high confidence for individual comparisons. We recommend using entropy as one component of a multi-metric diversity profile rather than as a standalone measure.

E Additional Experimental Results

E.1 Complete Cross-Model τ Matrix

Table 10: Complete pairwise Kendall's τ matrix for all 10 model families. Diagonal entries are 1.0 by definition. The matrix is symmetric.

	GPT-4.1n	GPT-4.1m	Llama-3	Mistral	Phi-3	GPT-2	Claude-3H	Gemma-2	Qwen-2	Yi-1.5
GPT-4.1n	1.00	.35	.31	.34	.30	.33	.35	.31	.32	.33
GPT-4.1m	.35	1.00	.33	.36	.32	.31	.33	.34	.35	.32
Llama-3	.31	.33	1.00	.38	.33	.30	.32	.35	.33	.34
Mistral	.34	.36	.38	1.00	.34	.32	.36	.33	.34	.31
Phi-3	.30	.32	.33	.34	1.00	.29	.31	.29	.30	.33
GPT-2	.33	.31	.30	.32	.29	1.00	.34	.32	.33	.30
Claude-3H	.35	.33	.32	.36	.31	.34	1.00	.33	.34	.35
Gemma-2	.31	.34	.35	.33	.29	.32	.33	1.00	.31	.32
Qwen-2	.32	.35	.33	.34	.30	.33	.34	.31	1.00	.34
Yi-1.5	.33	.32	.34	.31	.33	.30	.35	.32	.34	1.00

Key observations from the full matrix:

- The highest pairwise τ is 0.38 (Llama-3 vs. Mistral), suggesting that open-weight models with similar training approaches may share diversity characteristics.
- The lowest pairwise τ is 0.29 (Phi-3 vs. Gemma-2 and Phi-3 vs. GPT-2), indicating that smaller models have less predictable diversity profiles.
- The range of τ values ($[0.29, 0.38]$) is narrow, which explains the negligible difference between within-family and cross-family averages.

E.2 Per-Prompt Metric Variability

Table 11: Coefficient of variation (CV) of diversity metrics across models, by prompt type. Lower CV indicates more stable metrics.

Prompt Type	Distinct-2	Avg. PW	Entropy	Vendi
Open-ended creative	0.31	0.22	0.18	0.25
Factual question	0.15	0.11	0.09	0.13
Instruction-following	0.23	0.17	0.14	0.19
Summarization	0.12	0.09	0.08	0.11
Overall	0.22	0.16	0.13	0.18

Entropy is the most stable metric across models (lowest CV), while Distinct-2 is the most variable. Open-ended creative prompts yield the most variable diversity scores, as expected.

E.3 Selection Algorithm Scaling

All three algorithms improve with larger pools, but the heuristic methods benefit more, with the gap between random and farthest-point widening from 13.8 at $n = 50$ to 17.7 at $n = 500$.

Table 12: Selection algorithm spread as a function of n (pool size) with fixed $k = 20$.

n	Random	Farthest Point	Greedy Submodular
50	58.3 ± 5.1	72.1 ± 5.3	74.9 ± 3.1
100	60.1 ± 4.9	76.4 ± 4.9	79.3 ± 2.5
200	61.9 ± 4.9	78.7 ± 4.8	81.7 ± 2.1
500	63.2 ± 4.7	80.9 ± 4.5	83.8 ± 1.9

E.4 Theorem Verification: Distribution of Tightness Ratios

The tightness ratio $r = \frac{(n-1)\bar{d}^2}{2n\,s^2}$ has the following distribution across the 1,200+ test instances:

Table 13: Distribution of tightness ratios across all verified instances.

Percentile	5th	25th	50th	75th	95th	Max
r	0.42	0.63	0.75	0.85	0.96	0.999

The median ratio of 0.75 indicates that typical configurations use about 75% of the bound's capacity. The 95th percentile at 0.96 shows that many natural configurations approach the bound closely.

E.5 Adversarial Instance Analysis

Of the 200 adversarial instances:

- 79 instances (39.5%) are boundary cases ($r > 0.95$).
- 12 instances (6.0%) achieve $r > 0.99$ (near-tight).
- The remaining 121 instances have $r \in [0.50, 0.95]$.
- No instance achieves $r > 1.0$ (all satisfy the bound).

The adversarial instances were constructed using three strategies:

1. **Outlier-cluster:** One point far from a tight cluster of $n - 1$ points. This maximizes s relative to $\bar{d}$, so the bound holds trivially.
2. **Balanced bipolar:** Two clusters of $n/2$ points each, maximizing $\bar{d}$ relative to s. This is the hardest case for the bound.
3. **Simplex vertex:** Points placed at vertices of a regular simplex, which distributes mass uniformly.

F Lattice Structure Analysis

F.1 Metric Ordering as a Partial Order

Given a collection of generated sets $\mathcal{S} = \{S_1, \ldots, S_m\}$ and a set of diversity metrics $\mathcal{M} = \{D_1, \ldots, D_q\}$, each metric induces a total order on $\mathcal{S}$. The intersection of these total orders defines a partial order:

$$S_i \preceq S_j \;\Leftrightarrow\; D_k(S_i) \leq D_k(S_j) \;\; \forall\, k \in [q]. \tag{47}$$

Proposition F.1 (Lattice Structure)**.** *If the metric-induced partial order $(\mathcal{S}, \preceq)$ is a lattice, then for any two sets S_i, S_j, there exist unique meet $(S_i \wedge S_j)$ and join $(S_i \vee S_j)$ elements satisfying:*

$$D_k(S_i \wedge S_j) \leq \min(D_k(S_i), D_k(S_j)) \quad \forall\, k, \tag{48}$$

$$D_k(S_i \vee S_j) \geq \max(D_k(S_i), D_k(S_j)) \quad \forall\, k. \tag{49}$$

F.2 When Does Lattice Structure Hold?

In general, the metric-induced partial order is *not* a lattice. It is a lattice when:

1. All metrics are monotone transformations of a single underlying quantity (trivially).
2. The collection $\mathcal{S}$ is small enough that no "incomparable pairs" create violations.
3. Metrics are restricted to a single family (e.g., all distributional), in which case Corollary 4.4 guarantees compatible orderings.

In our experiments, the partial order is a lattice for the distributional metrics alone but *not* when surface and IT metrics are included. The fraction of comparable pairs (out of all $\binom{m}{2}$ pairs) is 0.67 for distributional metrics alone and 0.41 for the full metric set.

F.3 Practical Implications

The lattice analysis provides a principled way to identify "universally diverse" and "universally non-diverse" sets—those that are maximal or minimal in the partial order. DivFlow exposes this through:

```
# Find sets that are diverse by all metrics
universal = divflow.universally_diverse(
    sets=all_sets,
    metrics=divflow.default_metrics()
)
```

In our cross-model experiments, approximately 15% of generated sets are universally diverse (top quintile on all four metrics simultaneously) and 12% are universally non-diverse. The remaining 73% are metric-dependent, reinforcing the importance of multi-metric evaluation.

G Implementation Details

G.1 Software Architecture

DivFlow is organized as a Python package with the following module structure:

```
divflow/
    __init__.py
    metrics/
        __init__.py
        surface.py       # TTR, Distinct-n, Self-BLEU
        distributional.py # Pairwise distances, spread,
                         # Vendi score
        information.py   # Entropy, NMI, VI
        registry.py      # Metric registration and
                         # discovery
    selection/
```

```
        __init__.py
        random_select.py
        farthest_point.py
        greedy_submodular.py
    correction/
        __init__.py
        miller_madow.py
        laplace.py
        jackknife.py
    crossmodel/
        __init__.py
        comparison.py
        power_analysis.py
        bootstrap.py
    verification/
        __init__.py
        smt_verify.py
        ilp_verify.py
        theorem_check.py
    utils/
        __init__.py
        embeddings.py
        statistics.py
        visualization.py
```

G.2 Dependencies

DivFlow requires:

- Python $\geq$ 3.9
- NumPy $\geq$ 1.21
- SciPy $\geq$ 1.7
- scikit-learn $\geq$ 1.0
- sentence-transformers $\geq$ 2.2
- z3-solver $\geq$ 4.12 (optional, for SMT verification)
- PuLP $\geq$ 2.7 (optional, for ILP verification)

G.3 Computational Complexity

Here, L is the average text length in tokens.

G.4 Embedding Pipeline

The embedding pipeline supports both local sentence transformers and API-based embeddings. For local models, we use batched inference with automatic GPU detection:

```
from divflow.utils.embeddings import embed_texts

# Automatic batching and GPU usage
embeddings = embed_texts(
    texts=texts,
```

Table 14: Time complexity of key operations (n = number of texts, d = embedding dimension, k = subset size).

Operation	Time	Space	Parallelizable?
Embedding	$O(n \cdot L)$	$O(n \cdot d)$	Yes (batch)
Avg. pairwise	$O(n^2 d)$	$O(n^2)$	Yes
Spread	$O(nd)$	$O(nd)$	Yes
Distinct-k	$O(n \cdot L)$	$O(n \cdot L)$	Yes
Self-BLEU	$O(n^2 L)$	$O(nL)$	Yes
Vendi score	$O(n^2 d + n^3)$	$O(n^2)$	Partial
FP selection	$O(nkd)$	$O(nd)$	Partial
Greedy submod.	$O(nk^2 d)$	$O(nd)$	Partial
SMT verify	NP-hard	$O(n^2)$	No

```
    model="all-MiniLM-L6-v2",
    batch_size=64,
    normalize=True
)
```

Normalization to unit vectors is applied by default for cosine-distance metrics but can be disabled for Euclidean distance computations.

G.5 Statistical Utilities

The `statistics` module provides:

- **Bootstrap CI:** Percentile and BCa intervals with configurable resampling count (default: 10,000).
- **Clopper–Pearson CI:** Exact binomial intervals for pass-rate verification.
- **Power analysis:** Post-hoc power computation for Kendall's τ tests, using the asymptotic distribution of τ under the alternative.
- **Effect size:** Cohen's d, Hedges' g, and Glass's Δ.

G.6 Reproducibility

All random operations in DivFlow accept an optional `seed` parameter. When provided, results are deterministic across runs. The cross-model experiments in this paper used `seed=42` throughout.

Experiments were conducted on a machine with:

- CPU: AMD EPYC 7763 (64 cores)
- RAM: 256 GB
- GPU: NVIDIA A100 40GB (for embedding computation)
- OS: Ubuntu 22.04
- Python 3.11.5

Total compute time for all experiments reported in this paper: approximately 14 hours (of which 11 hours were embedding computation and 2.5 hours were SMT verification).

Additional References

References

[1] H. Callaert and P. Janssen. The Berry–Esseen theorem for U-statistics. *The Annals of Statistics*, 6(2):417–421, 1978.

[2] L. de Moura and N. Bjørner. Z3: An efficient SMT solver. In *Proceedings of TACAS*, 2008.

[3] R. J. Serfling. *Approximation Theorems of Mathematical Statistics*. Wiley, 1980.